

Casimir wormhole solutions in $f(R, \mathcal{L}_m)$ gravity

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Highlights

- The derived Casimir wormhole solutions are stable.
- Small quantity of exotic matter for stabilization.
- Violation of Null Energy Condition is obtained.

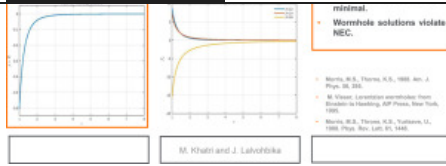
Abstract

In this paper, we study the Casimir effect on the wormhole geometry in $f(R, \mathcal{L}_m)$ gravity. We derive the field equations for the generic $f(R, \mathcal{L}_m)$ function by assuming static and spherically symmetric Morris–Thorne wormhole metric. Then we consider two non-linear $f(R, \mathcal{L}_m)$ models, specifically, $f(R, \mathcal{L}_m) = \frac{R}{2} + \mathcal{L}_m^\beta$ and $f(R, \mathcal{L}_m) = \frac{R}{2} + (1 + \alpha \mathcal{L}_m) \mathcal{L}_m$ where α and β are free parameters. We derive the shape functions for wormholes by utilizing the Casimir effect and examining their existence. Subsequently, we analyse the obtained wormhole solutions for each scenario, assessing the energy conditions at the wormhole throat with a radius of r_0 . Our findings reveal that for some arbitrary quantities, there is a violation of classical energy conditions at the wormhole throat. Additionally, we delve into the behaviour of the equation of state (EoS) for each case. Furthermore, we explore the stability of the Casimir effect wormhole solutions by employing the generalized Tolman–Oppenheimer–Volkoff (TOV) equation. Finally, we utilize the volume integral quantifier to determine the amount of exotic matter required near the wormhole throat for both models.

Graphical abstract

Article preview

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Keywords

Casimir effect; Wormhole; Modified gravity; $f(R, L_m)$ gravity; Morris–Thorne spacetime

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Introduction

In 1916, a year following Einstein's introduction of the Theory of General Relativity (GR), Schwarzschild presented the inaugural non-trivial analytic solution for Einstein's GR [1]. This solution by Schwarzschild exhibits a removable singularity at a regular point termed the event horizon. This concept laid the foundation for the idea of a black hole, whose existence has recently been confirmed [2], [3]. Concurrently, L. Flamm identified another potential solution, now recognized as a wormhole [4]. A wormhole is a geometric configuration that resembles a corridor connecting various universes or separate regions within the same universe. Subsequently, Einstein and Rosen delved into the precise nature of a wormhole, forming a hypothetical bridge based on Flamm's idea, now known as the Einstein–Rosen bridge [5]. Essentially, wormholes serve to link two asymptotically flat regions of space–time through a throat [6]. Finally, Morris and Thorne, in [7], introduced static traversable wormholes that redefine interstellar travel and offer exact solutions within general relativity (GR). The authors demonstrated that a wormhole could be traversable if it exhibits exotic matter with minimal surface area, satisfying the flare-out condition. Exotic matter, violating null energy conditions (NEC) and defined by $T_{\mu\nu}k^\mu k^\nu \geq 0$ for any null vector k^μ (where $T_{\mu\nu}$ is the stress–energy tensor), is considered an essential ingredient in constructing a wormhole in GR. Indeed, in classical GR, wormhole solutions contravene all energy conditions [6]. However, for a physically plausible model, challenging the existence of the hypothetical fluid becomes necessary. Since ruling out the presence of such a candidate within the GR framework was difficult [8], [9], an alternative approach gained support, aiming to minimize or even eliminate the dependence on exotic matter [10], [11], [12]. It is known that GR describes the universe well enough; however, some recent observations demand modifying the classical relativity into some alternative that could explain well, such as the late-time acceleration expansion of the universe [13], [14], [15], [16]. To tackle this, several modified theories are introduced such as $f(R)$ gravity [17], [18], $f(T)$ gravity [19], [20], $f(Q)$ gravity [21], Rastall gravity [22] etc.

in [24]. This discussion specifically omits exotic matter. The exploration of wormhole solutions has gained momentum, evidenced by Maldacena and Milekhin's creation of traversable WHs using the Randall–Sundrum model [25]. Additionally, Rahaman et al. put forward the notion of WH existence within the galactic halo [26]. Investigating blackbounce spacetime, studies have delved into the dynamics of spherically symmetric traversable WHs [27]. Garcia and Lobo analysed WH behaviour within curvature-matter coupling gravity [28]. The literature also encompasses WH solutions across various gravitational theories, including teleparallel gravity [29], extended theories of teleparallel gravity [30], [31], brane [32], Einstein–Gauss–Bonnet theory [33], Brans–Dicke theory [34], $f(R, T)$ gravity [35], and Rastall gravity [36].

Dutch physicist Hendrik Casimir proposed the Casimir effect in 1948, suggesting the existence of a force between uncharged, parallel, conducting plates [37]. Garattini [38] viewed Casimir's energy as a potential avenue for exploring traversable wormholes. The Casimir effect's strength is intricately tied to the geometry of the plates and serves as an artificial energy source. This energy form becomes particularly relevant for traversable wormholes, as the quantum field of the vacuum between parallel uncharged plates generates a negative energy density. Lamoreaux's experimental confirmation in 1997 [39] supported the earlier hypothesis about the Casimir effect, and subsequent refinements by Bressi [40] yielded even more precise results. In the context of the weak limit approximation, Javed et al. [41] delved into examining the weak deflection angle of photons around a Casimir wormhole. They applied the Gauss–Bonnet theorem to the Gaussian optical spacetime to determine the Gaussian optical curvature. In a separate publication, the authors investigated the connection between a remarkably harmless traversable wormhole and Casimir energy [42]. They extended the concept of an Absurdly Benign Traversable Wormhole, revealing that the wormhole throat is Planckian but significantly large. Muniz et al. conducted a study on the Casimir effect between parallel plates within the spacetime of a rotating wormholes [43]. Casimir wormhole have been extensively studied in several modified gravity theories [44], [45], [46].

Among the generalized $f(R)$ gravity, the theory that takes into account the influence of an 'extra' force by considering the coupling between the matter Lagrangian density and the Ricci Scalar is $f(R, \mathcal{L}_m)$ gravity [13], [47]. Interestingly, this additional force, perpendicular to the four-velocity, is accountable for the non-geodesic motion of the test particles. Wang and Liao derived the energy conditions in $f(R, \mathcal{L}_m)$ gravity [48]. In [49], Solanki et al. obtained wormhole solutions in $f(R, \mathcal{L}_m)$ gravity by assuming the linear barotropic, anisotropic, and isotropic equation of state (EoS). Also, Naseer et al. constructed a traversable wormhole solution in $f(R, \mathcal{L}_m)$ theory [50]. Several works have been done in the context of this theory [49], [51], [52], [53]. In the present work, we have explored the effect of the Casimir wormhole in the framework of $f(R, \mathcal{L}_m)$ gravity.

The present manuscript is organized as follows: Firstly, in Section 2, we review the $f(R, \mathcal{L}_m)$ gravity theory and obtain the field equations for the Morris–Thorne wormhole with an anisotropic matter background. Along with this, we review the Casimir effect between two parallel plates. Then in Section 3, we investigate the behaviour of the shape function and energy conditions for the Casimir wormhole

conclude our results in Section 6.

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Wormholes in modify gravity theory

In this section, we explore the basic formulation of gravity represented by $f(R, \mathcal{L}_m)$ and derive the corresponding governing field equation. Furthermore, we examine the generation of wormholes incorporating the Casimir effect within the context of this theoretical framework. ...

Wormhole solutions under the Casimir effect in $f(R, \mathcal{L}_m)$ gravity

For our study, we will consider two minimal $f(R, \mathcal{L}_m)$ functions which are motivated by the generic $f(R, \mathcal{L}_m)$ function, specifically,

$f(R, \mathcal{L}_m) = f_1(R) + f_2(R)G(\mathcal{L}_m)$ that represents arbitrary curvature-matter coupling [61]. To avoid the event horizon scenario, we assume $\varphi'(r) = 0$. Moreover, the matter Lagrangian is taken to be $\mathcal{L}_m = \rho$ [62]. ...

Equilibrium conditions

In this section, we investigate the stability of our obtained wormhole solutions for both models by using the generalized Tolman–Oppenheimer–Volkoff (TOV) equation [67], [68]. The generalized TOV equation can be defined as

$$-\frac{dP_r}{dr} - \frac{\varphi'(r)}{2}(\rho + P_r) - \frac{2}{r}(P_r - P_t) = 0.$$

The generalized Tolman–Oppenheimer–Volkoff (TOV) equation, denoted as Eq. (4.1), describes the equilibrium condition for a wormhole subjected to hydrostatic $\mathcal{F}_h$, gravitational $\mathcal{F}_g$, and anisotropic $\mathcal{F}_a$ forces. These forces include pressure from ...

Amount of exotic matter

symmetric spacetime with anisotropic matter distribution is given by

$$\Theta = \oint [\rho + P_r] dV,$$

where the volume can be read as $dV = r^2 dr d\Omega$ where $d\Omega$ is the solid angle. The VIQ can be rewritten as follows:

$$\Theta = \int_{r_0}^{\infty} [\rho + P_r] 8\pi r^2 dr,$$

Now, the volume integral corresponding to a ...

Conclusion

A wormhole remains a fascinating hypothetical spacetime structure, and its actual existence poses a significant question. In recent decades, the research community has actively delved into unravelling the nature of this mathematically defined geometric entity. Modified theories, rather than General Relativity (GR), have proven to be more influential in the exploration of wormholes. In this context, we investigate Casimir wormholes within $f(R, \mathcal{L}_m)$ gravity and derive modified motion equations for ...

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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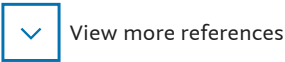
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